DELOOPING AS AN EMBEDDING

Purpose. Informal proof for `doc/delooping-embedding-handoff.txt`: looping is an embedding of pointed 2-connected 4-types into pointed 1-connected 3-types, the resulting description of the former inside the classification by triples, the delooping of $R = \mathrm{Fib}\bar{k}$, and the precise statement of the marked universal property (step 2 of the handoff) together with an analysis of what the embedding does and does not give for it.
Attribution. The organising principle (delooping is an embedding, and $\pi_4(S^3)$ is read off inside the deloopable part of the classification) and the sphere universal-property strategy are David Wärn's; the delooping machinery (`DeloopMaps`, DW1 §3) is his. The classification of pointed simply connected 3-types by triples is Hart–Milner's (Owen Milner's equivalence; `doc/postnikov-classification.pdf`).
Conventions. As in the development, $\mathsf{isConn}\,n\,X$ means that X is $(n-2)$-connected and $\mathsf{isTrunc}\,n\,X$ that X is an $(n-2)$-type. Write

$$\mathsf{SC3} = (X:\mathcal{U})\times(x:X)\times\mathsf{isConn}\,3\,X\times\mathsf{isTrunc}\,5\,X, \qquad \mathsf{SC4} = (Y:\mathcal{U})\times(y:Y)\times\mathsf{isConn}\,4\,Y\times\mathsf{isTrunc}\,6\,Y,$$

(pointed 1-connected 3-types, `ClassPost.SC3`; pointed 2-connected 4-types). Both are of the form $\mathsf{PP}_P = (X:\mathcal{U})\times(x:X)\times P\,X$ with P a family of propositions, so `ClassBase.PtProp` applies:

$$(u = v) \;\simeq\; \mathrm{PtEquiv}(u,v), \qquad (v:\mathsf{PP}_P)\times\mathrm{PtEquiv}(u,v)\ \text{is contractible}, \tag{1}$$

the first map being transport of id_u (the fundamental theorem of identity types applied to the second). Here $\mathrm{PtEquiv}(u,v) = (F : \mathrm{Map}_*(u,v))\times\mathsf{isEquiv}\,F$. All of $\mathsf{SC3}$, $\mathsf{SC4}$, $\mathsf{Triples}$ and the types below over them live in $\mathcal{U}_{\ell+1}$ (the carriers in $\mathcal{U}_\ell$), so the fundamental theorem, `sigma-base` and `emb-fib-prop` are used at level $\ell+1$; Tor_R below is in $\mathcal{U}_\ell$. Looping is

$$\Omega : \mathsf{SC4}\to\mathsf{SC3}, \qquad \Omega(Y,y,c,t) = (y = y,\ \mathsf{refl}_y,\ c',\ t'),$$

with $c' =$ `conn-id` and $t' = t\,y\,y$. For pointed maps $\mathsf{loop} : \mathrm{Map}_*(Y,Y') \to \mathrm{Map}_*(\Omega Y,\Omega Y')$ is `DeloopMaps.Maps.loop`; its fibre over f is (up to the symmetry of $=$) $\mathsf{Unloop}f = (F:\mathrm{Map}_*(Y,Y'))\times(f = \mathsf{loop}F)$.

1. Looping is an embedding

Lemma 1 (looping of maps)**.** *Let Y be pointed with $\mathsf{isConn}\,4\,Y$ and Y' pointed with $\mathsf{isTrunc}\,6\,Y'$. Then $\mathsf{loop} : \mathrm{Map}_*(Y,Y') \to \mathrm{Map}_*(\Omega Y,\Omega Y')$ is an equivalence.*

Proof. DW1 Corollary 11 (`Over.cor11`) at $c = 2$: the hypotheses are $\mathsf{isConn}\,(c+2)\,Y$ and $\mathsf{isTrunc}\,((c+1)+(c+1))\,Y'$, i.e. $\mathsf{isConn}\,4$ and $\mathsf{isTrunc}\,6$; then each $\mathsf{Unloop}f$ is contractible, and a map with contractible fibres is an equivalence. This is `SourceS3.LoopEquiv` at $c = 2$ (Lemma loopeq3 of `doc/br-universal.pdf`); its inverse sends f to the centre of $\mathsf{Unloop}f$, and $\mathsf{loop}(\text{centre}) = f$ is the second component. □

Lemma 2 (looping of equivalences)**.** *For $u = (Y,y,..)$, $v = (Y',y',..)$ in $\mathsf{SC4}$, write $\Omega_* : \mathrm{PtEquiv}(u,v) \to \mathrm{PtEquiv}(\Omega u,\Omega v)$ for $(F,e) \mapsto (\mathsf{loop}F,$ `loop-pe-equiv` $F)$ (`LoopIso.loopPE`). It has a section δ: $\Omega_*(\delta g) = g$ for every g.*

Proof. For $g = (f,e)$ let F be the inverse of Lemma 1 at f, so $\mathsf{loop}F = f$. Since $\mathsf{loop}F = f$ is an equivalence and Y, Y' are 0-connected ($\mathsf{isConn}\,2$, from $\mathsf{isConn}\,4$ by `conn-down`), F is an equivalence (Whitehead's principle, `Whitehead.Pointed.equiv`). Put $\delta g = (F,\ldots)$. Then

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$\Omega_*(\delta g) = (\mathsf{loop}F, \ldots) = (f, e)$: the first components agree by Lemma 1, the second ones lie in the proposition $\mathsf{isEquiv}\, f$. □

Only the section is used: a retract of a contractible type is contractible, so we never need that Ω_* is an equivalence, nor that it is compatible with ap_Ω under (1).

Theorem 3 (E)**.** *For every $u : \mathsf{SC4}$, the type $(v : \mathsf{SC4}) \times (\Omega u = \Omega v)$ is contractible. Consequently $\mathsf{ap}_\Omega : (u = v) \to (\Omega u = \Omega v)$ is an equivalence for all u, v (Ω is an embedding), and every fibre $\mathrm{fib}_\Omega(w) = (v : \mathsf{SC4}) \times (\Omega v = w)$, $w : \mathsf{SC3}$, is a proposition.*

Proof. Fix u. Consider the three total spaces over $v : \mathsf{SC4}$

$$T_1 = (v : \mathsf{SC4}) \times \mathrm{PtEquiv}(u, v), \quad T_2 = (v : \mathsf{SC4}) \times \mathrm{PtEquiv}(\Omega u, \Omega v), \quad T_3 = (v : \mathsf{SC4}) \times (\Omega u = \Omega v).$$

(1) T_1 is contractible by (1) for $\mathsf{SC4}$ (`PtProp.contr`).

(2) T_2 is a retract of T_1: $s(v, g) = (v, \delta g)$, $r(v, G) = (v, \Omega_* G)$, $r(s(v, g)) = (v, g)$ by Lemma 2. Hence T_2 is contractible.

(3) T_3 is a retract of T_2: by (1) for $\mathsf{SC3}$ at Ωu the map $\mathsf{tr}_v : (\Omega u = \Omega v) \to \mathrm{PtEquiv}(\Omega u, \Omega v)$ is an equivalence with inverse un_v; take $s(v, p) = (v, \mathsf{tr}_v p)$, $r(v, g) = (v, \mathsf{un}_v g)$. Hence T_3 is contractible.

By the fundamental theorem of identity types (`FundThm.FT` at the family $v \mapsto (\Omega u = \Omega v)$ and the point $\mathsf{refl}_{\Omega u}$), transport of $\mathsf{refl}_{\Omega u}$, $p \mapsto p_* \mathsf{refl}_{\Omega u}$, is an equivalence $(u = v) \to (\Omega u = \Omega v)$; it is ap_Ω (by J on p: both are refl at refl). An embedding has propositional fibres (`Groupoid.emb-fib-prop`). □

Remark 4. The proof uses about Ω only that it acts on PtEquiv through loop and that loop has a section on equivalences. The hypotheses $\mathsf{isConn}\, 4$, $\mathsf{isTrunc}\, 6$ enter only in Lemma 1 (and $\mathsf{isConn}\, 2$ in Whitehead's principle).

2. The deloopable triples

Recall $\mathsf{Triples} = (A\ B : \mathrm{Ab}) \times \mathrm{Map}_*(K(A,2), K(B,4))$ and the equivalence $\mathsf{realise} : \mathsf{Triples} \simeq \mathsf{SC3}$, $\mathsf{realise}(A, B, k) = (\mathrm{Fib}k, (\mathrm{pt}, k_0), ..)$ (`PostnikovClassification.classification`).

Definition 5. $\mathsf{Deloopable}\, t = \mathrm{fib}_\Omega(\mathsf{realise}\, t) = (v : \mathsf{SC4}) \times (\Omega v = \mathsf{realise}\, t)$.

Corollary 6 (C)**.** $\mathsf{Deloopable}\, t$ *is a proposition, and $(t, (v, p)) \mapsto v$ is an equivalence*

$$(t : \mathsf{Triples}) \times \mathsf{Deloopable}\, t \ \simeq\ \mathsf{SC4}.$$

Proof. A proposition by Theorem 3. The map is the composite of $(t, z) \mapsto (\mathsf{realise}\, t, z)$, an equivalence $(t : \mathsf{Triples}) \times \mathrm{fib}_\Omega(\mathsf{realise} t) \simeq (w : \mathsf{SC3}) \times \mathrm{fib}_\Omega(w)$ (Sigma over an equivalence of bases, `SphereAxiom.sigma-base`, as in `PostnikovClassification.base`), and $(w, (v, p)) \mapsto v$, $(w : \mathsf{SC3}) \times \mathrm{fib}_\Omega(w) \simeq \mathsf{SC4}$ (the total space of the fibres of a map is its domain; inverse $v \mapsto (\Omega v, (v, \mathsf{refl}))$, the composite on the other side by J on p). □

So pointed 2-connected 4-types are exactly the triples (A, B, k) together with the *property* that Fib$k$ deloops. In the classification a 4-type carries no data beyond its loop space.

3. Step 1: R deloops

Let $\bar{t} = (\mathbb{Z}, \mathbb{Z}/2, \bar{k})$ and $R = \mathrm{Fib}\bar{k} = \mathsf{realise}(\bar{t})$ (`DeloopR.TR`).

Proposition 7. $\mathsf{Deloopable}\, \bar{t}$*, witnessed by $BR = \mathrm{Fib}\hat{F}$ (`DeloopR`).*

Proof. (BR, br_0) lies in $\mathsf{SC4}$ by `BR-conn`, `BR-trunc`; `omega-BR` is a pointed equivalence $\Omega BR \simeq_* R$, which by (1) for $\mathsf{SC3}$ is a path $\Omega(BR) = \mathsf{realise}\, \bar{t}$. □

Corollary 8 (uniqueness of the delooping of R)**.** $BR \simeq_* \mathsf{Tor}_R$*, where Tor_R is the torsor delooping of `DeloopR` (Corollary 19 of `doc/cross-term.pdf`).*

Proof. Tor_R is $\mathsf{isConn}\,4$ and $\mathsf{isTrunc}\,6$ (`tor-conn`, `tor-trunc`) and $\Omega\mathsf{Tor}_R \simeq_* R$ (`tor-loops`); so it also inhabits $\mathsf{Deloopable}\,\bar{t}$. Two elements of a proposition are equal; the first projection gives $BR = \mathsf{Tor}_R$ in $\mathsf{SC4}$, and (1) for $\mathsf{SC4}$ a pointed equivalence. □

4. Step 2: the precise statement

Under the S^3 axiom (segment parameters $S^3, \mathrm{base}, \mathrm{surf}, \mathrm{ump}_3$), `SourceS3.Final` defines a class $s \in [S^3, BR]_*$ (no representative is chosen). For Y pointed write $s^* : [BR, Y]_* \to [S^3, Y]_*$, $c \mapsto s; c$ (`HomotopyClasses.compC`).

Statement 9 (M, the marked universal property of BR)**.** For every $(Y, y, ..) : \mathsf{SC4}$, $s^* : [BR, Y]_* \to [S^3, Y]_*$ is a bijection.

For $T = \|S^3\|_4$ (a pointed 2-connected 4-type, `Sphere3Trunc`) and BR this is exactly what step 3 consumes: the two-map Yoneda argument `Pi4Sphere.Two4` needs s^* bijective into $P = BR$ and into T, and gives $\pi_4 S^3 \cong \pi_4 T \cong \pi_4 BR \cong \mathbb{Z}/2$. Into BR it is (a) below directly ($BR = \mathrm{Fib}\hat{F}$ literally); so M is consumed only at T, and M for all of $\mathsf{SC4}$ is stronger than what is used.

What is proved.

(a) For $Y = \mathrm{Fib}K$, $K : K(A,3) \to_* K(B,5)$ any level-3 triple: `SourceS3.Final.rep-fib`; and along pointed equivalences, `rep-pteq`. No presentation is used.

(b) For every $Y : \mathsf{SC4}$: `Presentation4.Pres4` gives a level-3 triple K and $\mathrm{Fib}K \simeq Y$, pointed merely (Y is connected); M is a proposition, so (a) applies. This is `Final4.rep-T` with T replaced by an arbitrary Y.

Remark 10 (looped form; not used, not formalised). Let $e = \texttt{omega-BR} : \Omega BR \simeq_* R$ and

$$r = (\Omega s); [e] \;\in\; [\Omega S^3, R]_*,$$

a class (Ω acts on classes; s has no chosen representative). Since S^3 and BR are $\mathsf{isConn}\,4$ (`Sphere3Trunc.Simply.s3-conn`, `BR-conn`) and Y is $\mathsf{isTrunc}\,6$, Lemma 1 gives bijections $[BR, Y]_* \cong [\Omega BR, \Omega Y]_* \cong [R, \Omega Y]_*$ (the second by $[e]^{-1}$) and $[S^3, Y]_* \cong [\Omega S^3, \Omega Y]_*$, and looping commutes with composition and descends to classes (`Sphere3Trunc.loopC`, `loopC-comp`); so M at Y is equivalent to the bijectivity of

$$r^* : [R, \Omega Y]_* \longrightarrow [\Omega S^3, \Omega Y]_*, \qquad c \mapsto r; c.$$

By Corollary 6 the targets ΩY are exactly the $\mathsf{realise}\,t$ with t deloopable. The existing results give this at once for the t which are loopings of level-3 triples (looping (a)).

5. The crux: are all deloopable triples loopings?

For a level-3 triple (A, B, K) let

$$\Omega_3(A, B, K) = (A, B,\ e_2^{-1}; \mathsf{loop}K; e_4) \;:\; \mathsf{Triples},$$

with $e_2 : \Omega K(A,3) \simeq_* K(A,2)$, $e_4 : \Omega K(B,5) \simeq_* K(B,4)$ the Eilenberg–MacLane identifications, oriented as in the development (`H4Bilinear.Lev.eA`, `eB`; `DeloopR`: $f = e_2; \bar{k}; e_4^{-1}$). For $t = (A, B, k)$ put $f_t = e_2; k; e_4^{-1} : \Omega K(A,3) \to_* \Omega K(B,5)$.

Proposition 11. *For $t = (A, B, k) : \mathsf{Triples}$:*

(1) $\mathsf{Unloop}f_t$ *is a proposition, and it holds iff for every $p : \Omega K(A,3)$, $f_t = D_p$ (the translate of $|f_t|$ by p, `DeloopMaps.Maps.D`); no coherence at $p = \mathsf{refl}$ is needed.*

(2) $\mathsf{Unloop}f_t \to \mathsf{Deloopable}\,t$*: if $f_t = \mathsf{loop}K$ then* $\mathrm{Fib}K$ *deloops* $\mathsf{realise}\,t$.

(3) $\mathsf{Deloopable}\,t \to \mathsf{Unloop}f_t$ *holds if every pointed 2-connected 4-type is merely of the form* $\mathrm{Fib}K$ *(Pres4).*

Unconditionally, (1) and (2) with Theorem 3 give that $K \mapsto \mathrm{Fib}K$ *is an embedding* $\mathsf{Triples3} \hookrightarrow \mathsf{SC4}$ *(below); the Crux is exactly its surjectivity. With (3),* $\mathsf{Deloopable}\, t \simeq \mathsf{Unloop} f_t$ *and*

$$\mathsf{SC4} \;\simeq\; (t : \mathsf{Triples}) \times \mathsf{Unloop} f_t \;\simeq\; \mathsf{Triples3} = (A\ B : \mathrm{Ab}) \times \mathrm{Map}_*(K(A,3), K(B,5)),$$

the classification one level up, with the inverse $K \mapsto \mathrm{Fib}K$*.*

Proof. (1) DW1 Corollary 10 at $c = 2$, $t = 1$: $K(A,3)$ is $\mathsf{isConn}\,4$ and $K(B,5)$ is $\mathsf{isTrunc}\,7 = \mathsf{isTrunc}((2+1)+(2+1)+1)$, so each cell $(b : K(B,5)) \times C\,a\,b$ and $\mathsf{Unloop} f_t$ itself are propositions (`Over.Levels[2,1]`). If $f_t = D_p$ for all p, then $(b_0, |f_t|, p \mapsto \ldots)$ inhabits the cell at the base point, every cell is inhabited by connectedness (`conn-elim`), and `lemma9.r` gives $\mathsf{Unloop} f_t$: this is `DeloopR.unloop-contr` (cells `CrossTerm.Cross.cross-cell`; "no pointing in p is used"). Conversely $f_t = \mathsf{loop} F$ gives the cells by `lemma9`, and (b, h, c) in the cell at the base point gives $f_t = D_p$ after the singleton contractions of `Levels`. (`lemma9` is only a retract; the equivalence holds because both sides are propositions.)

(2) $\Omega\mathrm{Fib}K \simeq_* \mathrm{Fib}(\mathsf{loop}K)$ (`Fibres.omega-fib`) $\simeq_* \mathrm{Fib} f_t$ (along $f_t = \mathsf{loop}K$) $\simeq_* \mathrm{Fib}k$ (along e_2, e_4, from the square $f_t; e_4^{-1} = e_2^{-1}; k$): the proof of `DeloopR.omega-BR` with $(Z, \mathbb{Z}/2, \hat{F})$ replaced by (A, B, K). $\mathrm{Fib}K$ is $\mathsf{isConn}\,4$ and $\mathsf{isTrunc}\,6$ as in `BR-trunc` (`MapBR.Triple3.trunc6`) and `BR-conn`: for a general K the connectivity is `conn-from-point` with $\Omega\mathrm{Fib}K \simeq_* \mathrm{Fib}k$ and `Triple.conn3`, as in `BR-conn` (`Triple3` has no such field).

(3) $\mathsf{Unloop} f_t$ is a proposition, so we may take Y deloops $\mathsf{realise}\, t$ and $Y \simeq_* \mathrm{Fib}K$. Then $\mathsf{realise}\, t = \Omega Y = \Omega\mathrm{Fib}K = \mathsf{realise}(\Omega_3(A', B', K))$ by (2), so $t = \Omega_3(A', B', K)$ ($\mathsf{realise}$ is an equivalence); transporting, f_t becomes $e_2; (e_2^{-1}; \mathsf{loop}K; e_4); e_4^{-1} = \mathsf{loop}K$.

The embedding: $\Omega(\mathrm{Fib}K) = \mathsf{realise}(\Omega_3 K)$ by (2), $\mathsf{realise} \circ \Omega_3$ has propositional fibres ($\mathsf{realise}$ is an equivalence and the fibre of Ω_3 over t is $\mathsf{Unloop} f_t$ up to the contractible choice of k, a proposition by (1)), and Ω is an embedding; if $g \circ h$ and g are embeddings so is h.

For the last equivalence: $(t : \mathsf{Triples}) \times \mathsf{Unloop} f_t$ is $(A\,B) \times (k) \times (K) \times (f_{(A,B,k)} = \mathsf{loop}K)$; for fixed K, $k \mapsto f_{(A,B,k)}$ is an equivalence (pre- and post-composition with pointed equivalences), so $(k) \times (f_{(A,B,k)} = \mathsf{loop}K)$ is contractible. □

Status of the crux. The Crux below is mathematically established, through Pres4. What is missing is a proof that avoids that machinery. The embedding gives *uniqueness* of deloopings; it does not give their *existence* in the form of a level-3 presentation. Precisely, the statement which is proved (via Pres4) but for which a Pres4-free proof is open is

Statement 12 (Crux)**.** For every $t : \mathsf{Triples}$, $\mathsf{Deloopable}\, t \to \mathsf{Unloop} f_t$; equivalently (by Proposition 11 and Theorem 3) every $Y : \mathsf{SC4}$ is $\simeq_* \mathrm{Fib}K$ for a level-3 triple K (a genuine pointed equivalence, not merely: the triple is unique).

It is proved with `Presentation4.Pres4` (Proposition 11(3)). The remaining obligation for this route is a proof of the Crux independent of Pres4; we do not see one. By (1) it amounts to showing that f_t agrees with all its translates D_p (the cross term of k vanishes) as soon as $\mathrm{Fib}k$ is a loop space. Classically this holds because the cross term is the obstruction to k being an H-map, and an H-structure on $\mathrm{Fib}k$ over the addition of $K(A,2)$ forces it, the comparison $\mathrm{Fib}k \wedge \mathrm{Fib}k \to K(A,2) \wedge K(A,2)$ being highly connected; none of this (smash products, their connectivity, H-structures on fibres) is available in the development; one also needs the classical fact that the primitive classes in $H^4(K(A,2); B)$ are exactly the suspensions of $H^5(K(A,3); B) = \mathrm{Hom}(A/2, B)$ ("H-map ⇒ deloops"). An equivalent, more internal target: the cross term is a Whitehead square, which vanishes in an H-space. The only existing route to a level-3 K is Pres4's: the fibres of $Y \to \|Y\|_3$ carry a homogeneous structure (Lemma 4 over the simply connected Y), so $z \mapsto (E_z, \mu_z)$ is a map $\|Y\|_3 \to K(B,5)$ of torsors (`Classify` at $n = 2$). The embedding does not shortcut this: it compares deloopings of a *given* loop space, while the existence of a delooping of $\mathrm{Fib}k$ of the form $\mathrm{Fib}K$ is a statement about the cocycle k.

A narrower residual at t_T (suggested by the review; not checked in detail). At $A = \mathbb{Z}$: by (H4) (`H4Proof.h4`) every $k : K(\mathbb{Z},2) \to_* K(B,4)$ is merely $\kappa; K(g,4)$ for a unique $g : \mathbb{Z} \to B$, and $\mathsf{Unloop} f_t$ is a proposition, so a mere path suffices. If $2g = 0$ then $g = h \circ \rho$ and $k = \bar{k}; K(h,4)$, so $f_t = \mathsf{loop}(\hat{F}; K(h,5))$ deloops, by naturality of $K(h,-)$ and e_4 only. Hence at $(\mathbb{Z}, B, k)$ the Crux reduces to: *if* $\mathrm{Fib}(\kappa; K(g,4))$ *is a loop space, then* $2g = 0$ (the cross term of the cup square is $2g\,\iota \otimes \iota$). For t_T one needs in addition $\pi_3 T \cong \mathbb{Z}$, i.e. $\|T\|_3 \simeq K(\mathbb{Z},3)$ (Pres4 gives only an arbitrary $A = \Omega^3 \|T\|_3$). Neither input is the content of Pres4.

Consequences for the assembly. With (b), M holds for all of $\mathsf{SC4}$, and `Pi4Sphere` is unchanged (it uses M at T and BR). The Crux is a *sufficient* route to a Pres4-free assembly, not a demonstrated necessity. It is needed only at the single triple t_T of ΩT: if f_{t_T} deloops, to K say, then $T \simeq_* \mathrm{Fib} K$ by Theorem 3 and Proposition 11(2), a genuine pointed equivalence, and (a) gives M at T. A second route, which does not construct any delooping of a cocycle, is to prove M directly for deloopable targets, in the looped form: r^* bijective into $\mathsf{realise}\, t$ for every deloopable t. This is the route on which the embedding could genuinely simplify the proof, and it is the next thing to investigate. Our analysis does not show that replacing Pres4 needs another full presentation theorem.

6. What is formalised

Module `DeloopEmbedding` (spec `embedding.opens`; checked together with the whole development, with no deviation in any `.opens` spec):

- $\mathsf{SC4}$, Ω (`SC4`, `Acs`, `Omega`);
- Lemma 1: `SourceS3.LoopEquiv` at $c = 2$, used as `LoopEq.LE`;
- Lemma 2: `LoopEq.om`, `dl`, `dl-eq`, `el`, `delta`, `om-delta` (by definition), `fst-eq`, `loop-delta`;
- Theorem 3: `E.contr1`, `contr2`, `contr3`; `Emb.tr-ap`, `Emb.emb` (ap_Ω is an equivalence), `fib-prop`;
- Corollary 6: `Deloopable`, `deloopable-prop`, `tot-ret`, `tot-equiv`, `sc4-equiv` : $(t : \mathsf{Triples}) \times \mathsf{Deloopable}\, t \simeq \mathsf{SC4}$;
- step 1: `tbar`, `BR-pt`, `deloop-R`; `Tor-pt`, `deloop-Tor`, `BR-Tor` : $BR \simeq_* \mathsf{Tor}_R$;
- statement M and (b): `Mark[S3,base,surf,ump3].Marked`, `Mark.marked` (every $Y : \mathsf{SC4}$, through Pres4).

Not formalised: the looped form of M, Proposition 11, and the Crux independently of Pres4.